1-to'plam. *Yoy uzunligi – radius va markaziy burchak berilgan yoki izlanadi.*

1 Radiusi 9 sm, markaziy burchak 40°. Yoy uzunligini toping.

2 Radiusi 15 sm, markaziy burchak 72°. Yoy uzunligini toping.

3 Radiusi 8 sm, markaziy burchak 270°. Yoy uzunligini toping.

4 Radiusi 6 sm, yoy uzunligi 4π sm. Markaziy burchakni (daraja) toping.

5 Radiusi 10 sm, yoy uzunligi 5π sm. Markaziy burchakni toping.

6 Yoyning markaziy burchagi 50°. Yoy o'lchovi nechaga teng? Uzunlikni topish uchun yana qanday ma'lumot kerak?

2-to'plam. *Sektor yuzi – radius va markaziy burchak berilgan yoki izlanadi.*

1 Radiusi 6 sm, markaziy burchak 60°. Sektor yuzini toping.

2 Radiusi 12 sm, markaziy burchak 150°. Sektor yuzini toping.

3 Radiusi 9 sm, markaziy burchak 40°. Sektor yuzini toping.

4 Radiusi 20 sm, markaziy burchak 18°. Sektor yuzini toping.

5 Radiusi 10 sm, sektor yuzi 15π sm^2. Markaziy burchakni toping.

6 Radiusi 18 sm, sektor yuzi 27π sm^2. Markaziy burchakni toping.

3-to'plam. *Aralash: yoy uzunligi va sektor yuzi – bitta doira, bitta burchak. Ikkalasini aralashtirmang.*

1 Radiusi 6 sm, markaziy burchak 90°. Yoy **uzunligini** toping.

2 (Yuqoridagi doira, xuddi shu burchak.) Sektor **yuzini** toping.

3 Radiusi 9 sm, markaziy burchak 200°. Yoy **uzunligini** toping.

4 (Yuqoridagi doira, xuddi shu burchak.) Sektor **yuzini** toping.

5 Purkagich radiusi 18 fut bo'lgan maydonni 50° markaziy burchak bo'yicha sug'oradi. Purkagich uchi chizadigan yoyning uzunligini toping.

6 Xuddi shu purkagich sug'oradigan maydonning (sektor) yuzini toping.

4-to'plam. *Markaz (h, k) va radius r ni toping.*

1 $(x-2)^2+(y-5)^2=49$

2 $(x+3)^2+(y-1)^2=16$

3 $(x-7)^2+(y+6)^2=81$

4 $x^2+(y-4)^2=25$

5 $(x+1)^2+(y+1)^2=2$

6 $(x-5)^2+y^2=25$

5-to'plam. *To'liq kvadratga keltiring:* $(x-h)^2+(y-k)^2=r^2$. *Markaz va radiusni toping.*

1 $x^2+y^2-4x-6y-12=0$

2 $x^2+y^2+10x-4y+13=0$

3 $x^2+y^2-2x+8y+1=0$

4 $x^2+y^2+6x+6y+9=0$

5 $x^2+y^2-8x+2y-8=0$

6 $x^2+y^2+2x-12y+12=0$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 Radiusi 9 sm, markaziy burchak 40° bo'lgan yoyning uzunligini toping.

"Yoy uzunligi $=40$ *sm."*

Xato va to'g'ri javob: .....

2 Radiusi 10 sm, markaziy burchak 90° bo'lgan sektorning yuzini toping.

"Sektor yuzi $=\frac{90}{360}\times 2\pi\times 10=5\pi$ *sm²."*

Xato va to'g'ri javob: .....

3 $x^2+y^2-10x+4y-7=0$ markazi va radiusini toping.

"$(x^2-10x)+(y^2+4y)=-7$, $(x-5)^2+(y+2)^2=-7+25+4=22$."

Xato va to'g'ri javob: .....

4 $(x+6)^2+(y-3)^2=49$ markazini toping.

"Markaz $(6,3)$."

Xato va to'g'ri javob: .....

5 Radiusi 16 sm, yoy uzunligi 8π sm. Markaziy burchakni toping.

"$8\pi=\frac{\theta}{360}\times\pi\times 16^2$, *bundan* θ *ni topamiz."*

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin.*

1 Sektorning yuzi 24π sm^2, markaziy burchagi 60°. Doiraning radiusini toping.

2 Yoyning uzunligi 10π sm, markaziy burchagi 100°. Avval radiusni, so'ngra xuddi shu burchak uchun sektor yuzini toping.

3 $x^2 + y^2 - 14x - 8y + 40 = 0$. Markaz va radiusni toping, so'ngra $(7, 4)$ nuqtasi doira ichida, tashqarisida, yoki aylananing ustida ekanligini aniqlang.

4 Sektorning yuzi to'liq doira yuzining $\frac{1}{8}$ qismi. Markaziy burchakni toping. Agar radius 16 sm bo'lsa, yoy uzunligini toping.

5 $x^2+y^2+4x-10y-20 = 0$. Markaz va radiusni toping. So'ngra markaziy burchagi 72° bo'lgan sektorning yoy uzunligi va yuzini toping.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	2π sm	2	6π sm
	3	12π sm	4	120°
	5	90°	6	o'lchov 50°; uzunlik uchun radius kerak
2-to'plam	1	6π sm^2	2	60π sm^2
	3	9π sm^2	4	20π sm^2
	5	54°	6	30°
3-to'plam	1	3π ft	2	9π sm^2
	3	10π sm	4	45π sm^2
	5	5π fut	6	45π fut^2
4-to'plam	1	$(2, 5)$, r=7	2	$(-3, 1)$, r=4
	3	$(7, -6)$, r=9	4	$(0, 4)$, r=5
	5	$(-1, -1)$, r=$\sqrt{2}$	6	$(5, 0)$, r=5
5-to'plam	1	$(2, 3)$, r=5	2	$(-5, 2)$, r=4
	3	$(1, -4)$, r=4	4	$(-3, -3)$, r=3
	5	$(4, -1)$, r=5	6	$(-1, 6)$, r=5
6-to'plam	1. Yoy o'lchovini uzunlik deb yozgan. To'g'ri: $\frac{40}{360} \times 2\pi \times 9 = 2\pi$ sm.			
	2. Yuza o'rniga uzunlik formulasi qo'llangan. To'g'ri: $\frac{90}{360} \times \pi \times 10^2 = 25\pi$ sm^2.			
	3. Erkin hadning ishorasi o'zgartirilmagan. To'g'ri: $(x-5)^2 + (y+2)^2 = 36$; markaz $(5, -2)$, $r = 6$.			
	4. $(x+6)^2$ ishorasi teskari o'qilgan. To'g'ri: markaz $(-6, 3)$.			
	5. Uzunlik o'rniga yuza formulasi qo'llangan. To'g'ri: $8\pi = \frac{\theta}{360} \times 2\pi \times 16$, $\theta = 90^\circ$.			
7-to'plam	1. $r = 12$ sm			
	2. $r = 18$ sm; sektor yuzi $= 90\pi$ sm^2			
	3. Markaz $(7, 4)$, $r = 5$; nuqta aynan markaz – doira ichida.			
	4. $\theta = 45^\circ$; yoy uzunligi $= 4\pi$ sm			
	5. Markaz $(-2, 5)$, $r = 7$; yoy uzunligi $= \frac{14\pi}{5}$ sm, sektor yuzi $= \frac{49\pi}{5}$ sm^2			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

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